

VeCAD Elite 2026

Embedded System - ST Edge AI Group

10-day Project Diary

TinyML Occupancy Estimation Prototype

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10-day Project Record

From MCU bring-up to a validated data-collection pipeline

DAY 1 Foundation and System Design

Confirmed the MCU and development environment were working.

DAY 2 Component and I2C Planning

Completed the component list and identified the expected device addresses.

DAY 3 STM32 AI Workflow

Understood that training occurs on the computer while inference runs on the STM32.

DAY 4 MNIST Deployment Study

Gained practical familiarity with the full PC-to-MCU inference workflow.

DAY 5 Hardware Preparation and Soldering

Prepared the physical modules for individual electrical testing.

DAY 6 Sensor Bring-Up

Validated the basic drivers and confirmed live readings on the OLED and serial output.

DAY 7 BH1750 Fault Diagnosis

Confirmed that the original GY-302 did not acknowledge on either controller.

DAY 8 VL53L0X Validation

Obtained stable Status 0 measurements for close targets.

DAY 9 Full Hardware Integration

Detected BH1750 0x23, VL53L0X 0x29, OLED 0x3C and BME280 0x76 together.

DAY 10 Data Logger Quality Review

Completed an end-to-end path from STM32 sensors to saved CSV data.

Tasks

- Reviewed the NUCLEO-L476RG architecture, memory, clock and header pinout.
- Installed STM32CubeIDE and STM32CubeMX, then verified the board with a Blink test.
- Defined the system architecture and mapped the sensors, controls and indicators.

Learning Sources and Tools

- ST NUCLEO-L476RG documentation
- STM32CubeIDE and STM32CubeMX getting-started guides
- I2C addressing, voltage levels and pull-up fundamentals

Achievement

- Confirmed the MCU and development environment were working.
- Completed an initial circuit and pin-assignment plan for the occupancy prototype.

Problem and Engineering Decision

- Verified shared-bus addressing and 3.3 V / 5 V compatibility before assembly.

IMAGE / EVIDENCE PLACEHOLDER

Insert the Blink test, NUCLEO board photo or circuit schematic

Day 2 | Component and I2C Planning

Tasks

- Ordered the required sensors and supporting components.
- Studied the interfaces of BME280, BH1750, VL53L0X, PIR and SSD1306 OLED modules.
- Planned one shared I2C bus and dedicated GPIO connections.

Learning Sources and Tools

- Sensor datasheets and module pinouts
- I2C seven-bit addresses and HAL shifted addresses
- Tutorials for individual sensor bring-up

Achievement

- Completed the component list and identified the expected device addresses.
- Used the delivery period to reduce integration risk before wiring the hardware.

Problem and Engineering Decision

- Components required several days to arrive, so software and protocol research continued in parallel.

IMAGE / EVIDENCE PLACEHOLDER

Insert the component list, order summary or final pin map

Day 3 | STM32 AI Workflow

Tasks

- Studied TinyML and the role of STM32Cube.AI / X-CUBE-AI.
- Mapped the path from labelled sensor data to model training and embedded inference.
- Explored model analysis, memory reporting and C code generation in STM32CubeMX.

Learning Sources and Tools

- STM32Cube.AI workflow and documentation
- TensorFlow / Keras model training basics
- MCU memory and computation constraints

Achievement

- Understood that training occurs on the computer while inference runs on the STM32.
- Defined a future workflow for converting the occupancy model into embedded C code.

Problem and Engineering Decision

- The AI toolchain was new, so this day focused on terminology, model formats and deployment steps.

IMAGE / EVIDENCE PLACEHOLDER

Insert an X-CUBE-AI configuration or model-analysis screenshot

Day 4 | MNIST Deployment Study

Tasks

- Followed a reference MNIST handwritten-digit CNN project for STM32.
- Reviewed Keras training, .h5 import, X-CUBE-AI analysis and code generation.
- Compared its NUCLEO-F411RE setup with the target NUCLEO-L476RG platform.

Learning Sources and Tools

- Szu-Chi: Implement MNIST Classification on MCU
- TensorFlow / Keras CNN workflow
- Virtual COM Port testing on an MCU

Achievement

- Gained practical familiarity with the full PC-to-MCU inference workflow.
- Used the project as a learning reference, not as the final occupancy model.

Problem and Engineering Decision

- The reference used a different STM32 board and software versions, so direct reuse required adaptation.

IMAGE / EVIDENCE PLACEHOLDER

Insert the model graph, X-CUBE-AI Analyze page or test output

Tasks

- Inspected the delivered modules and identified headers that required soldering.
- Arranged soldering tools and assembled the sensor modules.
- Checked pin labels, continuity, supply polarity and header joints.

Learning Sources and Tools

- Through-hole soldering and continuity checks
- Module power and signal-pin verification
- Breadboard wiring practices

Achievement

- Prepared the physical modules for individual electrical testing.
- Reworked weak joints that could create intermittent I2C communication.

Problem and Engineering Decision

- The modules were not ready to plug in immediately, which added soldering and inspection time.

IMAGE / EVIDENCE PLACEHOLDER

Insert a soldering photo or continuity-test result

Day 6 | Sensor Bring-Up

Tasks

- Tested the SSD1306 OLED and BME280 through I2C.
- Verified UART output and used an I2C scanner to confirm device responses.
- Tested PIR inputs, push buttons, LEDs and the active buzzer as GPIO devices.

Learning Sources and Tools

- STM32 HAL I2C and GPIO functions
- Serial Monitor debugging at 115200 baud
- Isolated testing before system integration

Achievement

- Validated the basic drivers and confirmed live readings on the OLED and serial output.
- Confirmed the active buzzer logic: PB4 LOW produces sound and HIGH is silent.

Problem and Engineering Decision

- Resolved several pin, port and power mistakes by testing one module at a time.

IMAGE / EVIDENCE PLACEHOLDER

Insert the OLED readings, GPIO test or Serial Monitor output

Tasks

- Scanned the expected BH1750 addresses 0x23 and 0x5C on STM32 and Arduino.
- Checked 3.3 V supply, SDA/SCL idle levels, soldering and the ADDR connection.
- Compared the module with known-good I2C devices and replacement BH1750 boards.

Learning Sources and Tools

- Cross-platform fault isolation
- BH1750 raw-data to lux conversion
- Separating wiring, software and hardware faults

Achievement

- Confirmed that the original GY-302 did not acknowledge on either controller.
- Verified a replacement at 0x23 and obtained valid lux readings.

Problem and Engineering Decision

- Diagnosed a faulty module only after eliminating code, wiring, address and soldering causes.

IMAGE / EVIDENCE PLACEHOLDER

Insert the failed scan beside the replacement 0x23 / lux output

Tasks

- Integrated the ST VL53L0X API and STM32 HAL platform layer.
- Removed Windows-only communication files and corrected build errors.
- Logged distance, range status, signal rate and ambient rate.

Learning Sources and Tools

- VL53L0X API initialization and calibration
- RangeStatus interpretation
- Embedded build-error diagnosis

Achievement

- Obtained stable Status 0 measurements for close targets.
- Displayed live distance data on the OLED and Serial Monitor.

Problem and Engineering Decision

- Observed 8190 mm / Status 4 when the target was outside the practical range or narrow field of view.

IMAGE / EVIDENCE PLACEHOLDER

Insert a valid distance test and an 8190 mm / Status 4 example

Tasks

- Connected four I2C devices on the same bus and verified their addresses.
- Integrated two PIR sensors, two buttons, two LEDs and the active buzzer.
- Added EMPTY, LOW and HIGH labels with button and LED feedback.

Learning Sources and Tools

- Multi-device I2C integration
- Button debounce and non-blocking buzzer control
- System-level power and wiring checks

Achievement

- Detected BH1750 0x23, VL53L0X 0x29, OLED 0x3C and BME280 0x76 together.
- Ran all sensors and user controls in one STM32 application.

Problem and Engineering Decision

- A weak supply/wiring connection initially hid the BME280, then a corrected connection restored the full bus.

IMAGE / EVIDENCE PLACEHOLDER

Insert the assembled prototype or four-device I2C scan

Day 10 | Data Logger Quality Review

Tasks

- Built a serial CSV logger with labelled EMPTY, LOW and HIGH sessions.
- Used physical controls to select labels and start or stop recording.
- Ran a 206-row pilot and applied data-quality checks.

Learning Sources and Tools

- Structured data collection and session labelling
- Raw-data preservation and validation flags
- Sensor suitability assessment before AI training

Achievement

- Completed an end-to-end path from STM32 sensors to saved CSV data.
- The first review accepted 44 rows and rejected 162, mainly because of invalid ToF measurements.

Problem and Engineering Decision

- PIR reports motion rather than continuous presence, while the current VL53L0X range and field of view do not cover a ceiling-mounted room setup.

IMAGE / EVIDENCE PLACEHOLDER

Insert the CSV file, pilot summary or sensor-validity chart

Final Achievement

A working integrated prototype with an evidence-based plan for the next iteration

Completed

- Brought up the NUCLEO-L476RG and implemented the required I2C and GPIO drivers.
- Integrated environmental, light, distance and motion sensing with OLED feedback.
- Built labelled session controls and a PC logger that saves structured CSV data.

Validated by Evidence

- Four devices detected on one I2C bus.
- Live values displayed and transmitted at 115200 baud.
- A 206-row pilot exposed data-quality limits before model training.

IMAGE / EVIDENCE PLACEHOLDER

Insert the final prototype photo or system block diagram

Next Stage

- Calibrate and reposition the two PIR modules, then add a motion-window feature in firmware.
- Use VL53L0X only for a close fixed zone, or evaluate a multizone ToF sensor for ceiling coverage.
- Repeat the pilot, collect balanced sessions, train the model and deploy it with STM32Cube.AI.